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ORIGINAL ARTICLE

Position information encoding FPN for small object detection in aerial images

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Abstract
Small object detection in aerial images is a challenge in remote sensing. Recently, convolutional neural networks (CNNs) have succeeded by learning localized filters that embed relative spatial information but fail to detect the small objects in the aerial images for the uneven padding. Even though the padding mechanism in CNNs allows for the capture of absolute position information and ensures consistent input–output resolution, it leads to a diminishing extraction of absolute position context from the edges to the center. This results in asymmetry bias, which negatively impacts position-dependent visual tasks like small object detection, causing blind spots and misdetection. In this study, we uncover that complex-valued CNNs, capable of explicitly encoding absolute position information, can significantly enhance conventional object detection techniques. To accomplish this, we introduce the position information encoding feature pyramid network (PieFPN), which consists of a complex-valued encoder and a real-valued decoder for explicit position information encoding. Additionally, we present the general Gaussian normalization and Gaussian error linear unit for multi-variables, incorporating them into end-to-end training schemes. To utilize the ImageNet pre-trained weights, we merge PieF with traditional feature pyramid networks, allowing for seamless integration into existing object detection pipelines. Our complex-valued designs outperform their real-valued counterparts, achieving state-of-the-art results on the DOTA-v2.0 oriented object detection in aerial images dataset.

Keywords Small object detection · Position information encoding · Convolutional neural networks · Feature pyramid network

1 Introduction

Object detection in aerial images has a wide range of applications in the real world, e.g., precision agriculture, disaster response, and economic activity [3, 21, 28]. However, the current object detection methods will fail to detect the small objects, which are abundant in aerial images. The major impact factor is the uneven padding mechanism CNNs, which leads the CNNs to forget the position information captured from the border of the input

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image [2, 18, 19]. Recent advancements in position information encoding have made significant progress in capturing the interdependence between tokens at various positions within input sequences for transformer-based methods. These innovations encompass absolute position embedding (APE) [11, 39, 41] and relative position embedding (RPE) [6, 25, 33, 36]. APE methods encode distinct input token positions in a sequence and merge them with input tokens to model positional information. In contrast, RPE methods determine token positions based on their relative distance to other tokens, learning pairwise relation embeddings through a look-up table within the self-attention mechanism. Owing to shift invariance and extrapolation, RPE enables transformers to capture long-range dependencies between tokens, surpassing APE in natural language processing (NLP) tasks [29, 42].

However, unlike NLP, which predominantly depends on relative position representations, recent research in computer vision (CV) reveals that CNNs not only learn relative position information in local filters within finite spatial receptive fields but also implicitly encode absolute position information, which is derived and propagated across the image from borders as prior anchors during spatial abstraction, is crucial for position-dependent visual tasks [18]. This combined encoding provides CNNs with information about the object content and its position in the image. Further studies suggest that absolute position information embedding allows CNNs to learn more distinguishable semantic features, leading to more accurate and confident predictions [19].

Consequently, the effective embedding of position information in CNNs has attracted growing interest. Two main approaches to absolute position information embedding in CNNs are implicit and explicit position encoding. The most prevalent implicit position encoding employs zero padding operations to introduce absolute spatial cues into CNNs [18, 19, 35]. Another implicit position encoding method enlarges the convolution kernel size to approximate global absolute position encoding [10]. However, due to the uneven padding mechanism in CNNs, the boundary effect and implicit position information recede from the borders toward the center, resulting in asymmetrical weight learning, which adversely affects position-dependent visual tasks like small object detection and generative tasks [2, 48]. To mitigate the negative impact of uneven padding in CNNs, researchers have proposed methods such as MS-PIE [48], which utilizes sinusoidal position encoding (SPE) [39] to explicitly encode absolute spatial information, guaranteeing a balanced spatial inductive bias throughout the image space.

In contrast to MS-PIE [48], we explore the ability of complex-valued neural networks (CVNNs) to explicitly encode both the global absolute and local relative spatial information in complex numbers. From a biological point of view, the complex-value formulation expresses the firing rate of the output of neurons as the amplitudes and the relative timing of their activity as the phases [34]. The synchronous input neurons with similar phases add constructively, whereas asynchronous neurons add destructively and thus interface with each other. In our work, the semantic context of the object is regarded as the amplitude and its position information as the phase. The complex neurons will aggregate both content and spatial information through complex convolutional operations. We present a novel position information encoding feature pyramid networks (PieFPN) for seamless integration into existing pipelines. PieFPN is the extension of standard feature pyramid networks (FPN) [24], adding a position information encoding fields (PieF) upon each output layer of FPN. Each PieF consists of a complex-valued encoder and a real-valued decoder. We apply multivariate Gaussian normalization and multivariate Gaussian error linear unit to train the complex-valued encoder in an end-to-end training scheme. We perform a sanity check of our PieF and demonstrate its effectiveness on oriented object detection in aerial images benchmark DOTA-v2.0 [9]. Our contributions in this paper are as follows:

1. We propose a position information encoding feature pyramid networks, which allows seamless integration into existing pipelines for utilizing the ImageNet [7] pre-trained weights and achieving fast convergence. (Sect. 3.2)
2. We extend the univariate normalization to the multivariate version, named multivariate Gaussian normalization. (Sect. 3.3)
3. We extend the univariate Gaussian error linear unit (GELU)[16] to the multivariate version, named multivariate Gaussian error linear uUnit. (Sect. 3.4)
4. A state-of-the-art result on DOTA-v2.0 oriented object detection in aerial image dataset. (Sect. 4.2)

2 Related work

Position information encoding in CNNs. CNNs are commonly regarded as translation equivariant and spatially agnostic, which means that the objects can appear anywhere in the image. However, recent researches show that CNNs enable to exploit absolute position information despite the pooling operation [2, 18, 19, and 35]. This absolute position information is implicitly disclosed by the zero padding, which is widely used for keeping the same resolution when adopting convolution operations [18]. Nevertheless, Alsallakh et al. [2] suggest that the uneven padding mechanism can induce spatial asymmetry bias in

CNNs, which is destructive to small object detection, e.g., traffic lights. Additionally, Xu et al. [48] reveal that zero padding will generate an unbalance inductive bias. This inductive bias is structured at the image edges and corners. At the same time, it is unstructured in the center region, since the effects of zero padding gradually diminish from borders toward the center.

To mitigate this unbalance spatial bias, some researchers propose using a padding of size two instead of a padding of size one in CNNs [2, 19, 35]. Semih Kayhan and van Gemert [35] find that a padding of size two allows CNNs to perform an equal number of convolution operations on all pixels, which is data efficiency and robustness to image shifts. Islam et al. [19] further reveal that a padding of size two allows CNNs to capture more global absolute spatial information. Unlike previous works, Ding et al. [10] scale up the convolution kernel size to approximate the global modeling.

In contrast to implicit position information encoding, Xu et al. [48] present the strengths of explicit position information encoding, including stable spatial inductive bias, and propose a multi-scale position information encoding training strategy by adding SPE [39] to inputs. Different from the aforementioned studies, we focus on exploring the ability of CVNNs to explicitly encode the position information by multiplying spatial embedding with the inputs and performing a sanity check to demonstrate its effectiveness. Wang et al. [41] propose a similar idea by training a complex-valued word embedding presenting the position bias for each token in the sequence for NLP tasks. In contrast, we design a novel position information encoding feature pyramid networks to explicitly model position information in 2D images.

Complex-valued neural networks. Complex numbers have a richer representation and facilitate noise-robust memory retrieval mechanisms. Regarding the history of CVNNs, Goto [12] first introduced phase information in computation, utilizing the phase of a high-frequency carrier to represent binary or multivalued information. In the present, CVNNs researchers extend the world of computation to pattern processing based on the structure of phase and amplitude information Reichert and Serre [34]. Trouillon et al. [38] handle the link prediction problem in NLP with complex-valued embedding to process various binary relations and automatically understand the structure of large knowledge bases. Wisdom et al. [45] utilize unitary recurrent matrices to address the vanishing and exploding gradient problems of recurrent neural networks. Recently, some researchers have tried to interpret the neural networks in terms of quantum physics on the semantic Hilbert space defined on a complex vector space [22, 40]. In CV, Worrall et al. [46] propose harmonic networks to capture the translation and rotation equivariance feature by learning radial and phase functions. Trabelsi et al. [37] present a general formulation for developing deep CVNNs, including complex batch normalization, complex weight initialization, and complex-valued ReLU-based activation function.

However, the aforementioned studies replace the whole pipeline with complex-valued operations and need to train with long epochs to achieve convergence. In contrast, we propose a strategy of combing real and complex representations in the feature pyramid networks, which allows seamless integration into existing pipelines for utilizing the ImageNet [7] pre-trained weights and achieving fast convergence. Our module consists of a complex-valued encoder and a real-valued decoder. Additionally, we present general multivariate Gaussian normalization and multivariate Gaussian error linear unit for developing the complex-valued encoder. The basic idea is also applicable to hypercomplex processing [5, 20, 32].

Object detection in aerial images. Object detection, the fundamental task in computer vision, is widely applied in many domains. For earth observation, object detection in aerial images has attracted many researchers' attention, which is possible to obtain accurate, current, and efficient information on precision agriculture, disaster response, and economic activity [3, 21, 28]. Unlike natural images, the objects in aerial images are in arbitrary orientation due to the overhead view. To handle rotation variations, Han et al. [14] present an anchor-based detector, ReDet, adopting group convolution [43] to extract the rotation equivariant features. R3Det [49] re-encodes the bounding boxes into the corresponding points. RoI Trans [8] learns the RoI-wise spatial transformation under the supervision of oriented bounding boxes (OBBs). In addition to anchor-based detectors, Guo et al. [13] propose an anchor-free detector, CFA, optimizing the feature assignment through convex-hull feature adaptation. Ori-RepPoints [23] predicts key points that bound the objects. In addition to being under the supervision of OBBs, H2RBox [50] and H2RBox-v2 [51] present an annotation-efficiency weakly supervised training pipeline for learning oriented representations from horizontal bounding boxes (HBBs).

Additionally, small object detection is a common problem in aerial images. In DOTA-v2.0 [9], annotated instances with a 10-50 pixels height occupy 77%. As shown before, zero padding will introduce an unbalance spatial bias in the learned weights, which is destructive to small object detection [2]. In this work, we propose PieF to explicitly encode the spatial information in complex numbers and mitigate the negative effect of implicit spatial information from the zero padding. And we show our approach achieves state-of-the-art on DOTA-v2.0. Next

section, we introduce the basic concept of CVNNs first and go on to our approach.

3 Position information encoding fields

3.1 Complex-valued neural networks

Consider a complex-valued neuron with the convolution kernel $W = A + iB \in \mathbb{C}$, where i is the imaginary unit, and given an input signal $z = x + iy \in \mathbb{C}$, the neuron generates a complex-valued output $W * z = (A * x - B * y) + i(B * x + A * y)$. Using matrix notation to present real and imaginary parts of the neuron output we have:

$$\begin{bmatrix} \Re(W * z) \\ \Im(W * z) \end{bmatrix} = \begin{bmatrix} A & -B \\ B & A \end{bmatrix} * \begin{bmatrix} x \\ y \end{bmatrix}. \tag{1}$$

The decision boundary is the different nature between a complex-valued neuron and a real-valued one, where a decision boundary of a single complex-valued neuron consists of two orthogonal hypersurfaces, and it divides a decision region into four equal sections [30]. Hence, the complex-valued neuron has richer representations than the real-valued one. However, the optimization of CVNNs encounters a problem, since the standard complex derivative exists only for holomorphic functions, i.e., satisfying the Cauchy–Riemann equations. The CVNNs commonly bring in nonholomorphic functions in two ways, including the loss function and activation functions. In such a scenario, derivative-based optimization approaches cannot use the standard complex derivatives. To deal with nonholomorphic functions, Wirtinger [44] observed that even if $f(z)$ isn't holomorphic, while $f(z, z^*)$ is always holomorphic, where z^* denotes the conjugate of z. And real and imaginary components of z can be expressed in terms of z and z^* as $\Re(z) = \frac{z+z^*}{2}, \Im(z) = \frac{z-z^*}{2i}$.

Hence, Wirtinger calculus provides an advanced tool for extending the well-known optimization algorithms in real-valued neural networks to CVNNs. Indeed, we implement the complex-valued encoder with the deep learning framework, Pytorch [31], and optimize it under the Wirtinger calculus.

3.2 Position information encoding feature pyramid networks

To utilize the ImageNet [7] pre-trained weights for fast convergence, we propose a feasible strategy by replacing the 3×3 feature fusion convolution layer with Position Information Encoding Fields (PieF), shown in Fig. 1 on page, which is seamless integration into existing object detection approaches. Next, we will detail the design of PieF.

Classical field theory, in terms of the physical, is the assignment of a physical quantity at each point of space and time, e.g., the wind velocity is described by a vector, representing the direction of the motion of air at each point in the space. All wind vectors in an area at a given point in time constitute a vector field. Borrowing ideas from classical field theory, we propose the position information encoding fields, which adopt the complex vector to represent the pixel position in the images.

We revisit the position information encoding in real domains. Generally, a 2D position information embedding in real domains is defined by a map $f^{\mathbb{R}}_{pie} : \mathbb{N}^2 \to \mathbb{R}^D$. The final embedding for each pixel is constructed by the sum $f(x, y) = f_\theta(x, y) \oplus f^{\mathbb{R}}_{pie}(x, y) \in \mathbb{R}^D$, where θ denotes the FPN Backbone and $\oplus$ is elementwise addition. Recently, SPE has been widely applied in NLP and 3D vision [27, 39]. Given $\omega_k = 1/10{,}000^{2k/d}$ (d presents half of the total embedding dimension, i.e., $d = D/2$) and a coordinate (x, y) in the 2D image, Xu et al. [48] present a 2D SPE as:

$$\begin{bmatrix} \underbrace{\left[\sin(\omega_0 y), \cos(\omega_0 y), \ldots, \sin(\omega_{\frac{d}{2}} y), \cos(\omega_{\frac{d}{2}} y)\right]}_{height}^T \\ \underbrace{\left[\sin(\omega_0 x), \cos(\omega_0 x), \ldots, \sin(\omega_{\frac{d}{2}} x), \cos(\omega_{\frac{d}{2}} x)\right]}_{width}^T \end{bmatrix}. \tag{2}$$

For any $z \in \mathbb{C}$, we obtain $z = re^{i\theta} = r(\cos\theta + i\sin\theta)$. And we can rewrite (2) into a complex format. To keep the same shape as the input, we utilize $\omega_k = 1/10{,}000^{k/d}$ and obtain the 2D SPE in complex domains $f^{\mathbb{C}}_{pie}$:

$$\left[\underbrace{e^{i\omega_0 y}, \ldots, e^{i\omega_d y}}_{height}, \underbrace{e^{i\omega_0 x}, \ldots, e^{i\omega_d x}}_{width}\right]^T. \tag{3}$$

Compared with (2), (3) can generate richer representations within the same quantity of elements. To achieve the final embedding, we merge the complex SPE and the backbone input by elementwise multiplication $\otimes$, obtaining $f(x, y) = f_\theta(x, y) \otimes f^{\mathbb{C}}_{pie}(x, y) = f_\theta(x, y) \otimes e^{i\omega(x,y)} \in \mathbb{C}^D$, which treats f_θ as the amplitude and $f^{\mathbb{C}}_{pie}$ as the phase. As shown in Fig. 1 on page 6, given a backbone input f_θ, we inject the position information $f^{\mathbb{C}}_{pie}$ twice. For each time, we adopt a complex encoder and real decoder to generate the enhanced feature representations $\mathcal{F}$. We utilize two MLPs to fusion $\mathcal{F}$ and f_θ. The architectures of the complex encoder and the real decoder are shown in Fig. 2 on page 7.. The complex

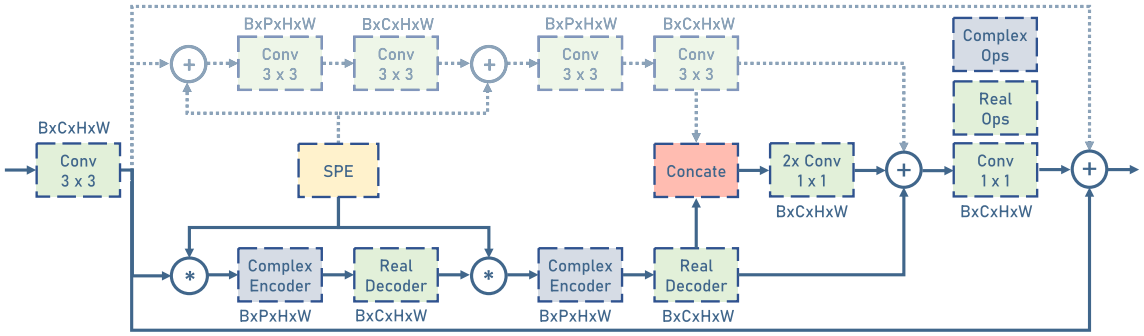


Fig. 1 Position information encoding fields. B = Batchsize, C = Output Channels of FPN, H = Height of the feature map, W = Width of the feature map, P = PieF Channels (To control the model size). The dashed pipeline denotes the real SPE. The solid pipeline represents our final method with complex SPE. Concatenation will be replaced by identity connection when only one pipeline works. We test different position information encoding methods. The comparison results are shown in Table 3 on page 12

encoder consists of a 3×3 complex convolution layer to aggregate the features from the local receptive field. To train the complex encoder, we adopt a complex group normalization (GN) [47] layer and a complex Gaussian error linear unit (GELU) [16] layer to accelerate convergence and avoid overfitting. In the real decoder, we concatenate real and imaginary parts of the complex features generated by the complex encoder in the complex to real module (CToR) first and adopt a 3×3 real convolution layer to decode the concatenated features in real domains, which is normalized by a GN layer and activated by a GELU layer. We will introduce the multivariate Gaussian normalization and multivariate Gaussian error linear unit in the following subsections and give special examples in complex numbers.

3.3 Multivariate Gaussian normalization

Deep neural networks generally rely on normalization to accelerate convergence [4, 17, 47].

However, univariate Gaussian normalization is unsuitable for complex numbers, since it does not ensure equal variance in both real and imaginary parts [37]. To handle this problem, Trabelsi et al. [37] propose a complex batch normalization ($\mathbb{C}BN$), which is a particular case of multivariate Gaussian normalization. Nevertheless, Ba et al. [4] and Wu and He [47] point out that the error of batch normalization rises rapidly when the batch size becomes smaller, caused by inaccurate batch statistics estimation. Hence, CVNNs also need the complex variant of layer normalization ($\mathbb{C}LN$) and group normalization ($\mathbb{C}GN$) to process the inaccurate batch statistics problem. Following the ideas of previous works, we present the general Multivariate Gaussian Normalization, which is applicable to CVNNs and hypercomplex processing [5, 20, and 32].

Consider an n-dimensional random vector $X = [X_1, X_2, \ldots, X_n]^T$, with mean $\mu_i = \mathbb{E}(X_i)$ and covariance matrix $\Sigma_{ij} = \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)] = Cov(X_i, X_j)$, if Σ is positive-define, the density function is given by

$$p(X; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(X-\mu)^T \Sigma^{-1}(X-\mu)\right). \tag{4}$$

The normalized vector is given by $\widetilde{X} = |\Sigma|^{-1/2}(X - \mu)$. Without loss of the generality, we discuss the relationship between bivariate Gaussian normalization and complex normalization, which is easy to promote to a high dimensional random vector and hypercomplex. In the bivariate nonsingular case ($rank(\Sigma) = n = 2$), given the correlation $\rho = Cov(X, Y)/\sqrt{Var(X)Var(Y)}$ and $\sigma_X > 0, \sigma_Y > 0$, the density function of a vector $[X, Y]^T$ is

$$p(X, Y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left\{-\frac{\left(\frac{(x-\mu_X)^2}{\sigma_X^2} - 2\rho\frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \frac{(y-\mu_Y)^2}{\sigma_Y^2}\right)}{2(1-\rho^2)}\right\}, \tag{5}$$

where the μ and Σ are formulated as:

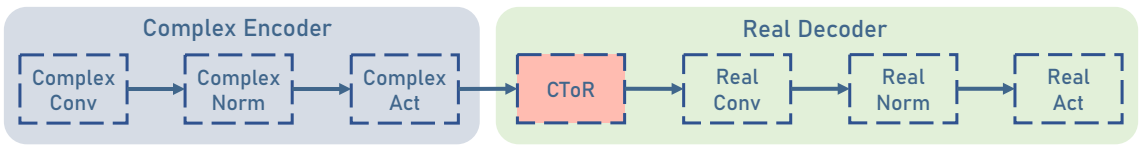


Fig. 2 Architecture of complex encoder and real decoder

$$\mu = \begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix}, \Sigma = \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}. \tag{6}$$

For complex normal distributions, consider the complex random vector $Z = X + iY$, with the mean $\mu = \mathbb{E}(Z) = \mathbb{E}(X) + i\mathbb{E}(Y)$, the covariance matrix $\Gamma = \mathbb{E}((Z-\mu)(Z-\mu)^H)$, and the relation matrix $C = \mathbb{E}[(Z-\mu)(Z-\mu)^T]$, the density function can be computed as

$$p(Z) = \frac{1}{\pi\sqrt{\det(\Gamma)\det(P)}} \exp\left\{-\frac{1}{2}\begin{pmatrix} \bar{Z}-\bar{\mu} \\ Z-\mu \end{pmatrix}^T \begin{pmatrix} \Gamma & C \\ \bar{C} & \bar{\Gamma} \end{pmatrix}^{-1} \begin{pmatrix} Z-\mu \\ \bar{Z}-\bar{\mu} \end{pmatrix}\right\}, \tag{7}$$

where $P = \bar{\Gamma} - C^H\Gamma^{-1}C$. The matrices Γ and C are related to covariance matrices of $[X, Y]^T$ via expressions:

$$\Gamma = \Sigma_{XX} + \Sigma_{YY} + i(\Sigma_{YX} - \Sigma_{XY}), \tag{8}$$

$$C = \Sigma_{XX} - \Sigma_{YY} + i(\Sigma_{YX} + \Sigma_{XY}). \tag{9}$$

The standard complex normal is with $\mu = 0, \Gamma = 1, C = 0$, i.e., $\mu_X = \mu_Y = 0, \Sigma_{XX} = \Sigma_{YY} = \frac{1}{2}, \Sigma_{XY} = \Sigma_{YX} = 0$. We can rewrite the (5) and (7) as:

$$p(X, Y) = \frac{1}{\pi}\exp\{-(X^2 + Y^2)\}, \tag{10}$$

$$\begin{aligned} p(Z) &= \frac{1}{\pi}\exp\{-(ZZ^*)\} \\ &= \frac{1}{\pi}\exp\{-(X^2 + Y^2)\} = p(X, Y). \end{aligned} \tag{11}$$

This provides a feasible way to normalize the complex random vector utilizing the univariate Gaussian normalization:

$$\widetilde{\Re(Z)} = \frac{\Re(Z) - \mathbb{E}(\Re(Z))}{\sqrt{2Var(\Re(Z))}} \sim \mathcal{N}\left(0, \frac{1}{2}\right), \tag{12}$$

$$\widetilde{\Im(Z)} = \frac{\Im(Z) - \mathbb{E}(\Im(Z))}{\sqrt{2Var(\Im(Z))}} \sim \mathcal{N}\left(0, \frac{1}{2}\right). \tag{13}$$

It allows us to apply state-of-the-art normalization technologies to the real and imaginary part of complex numbers directly, e.g., LN and GN. To promote to the hypercomplex, we obtain:

$$\widetilde{X_i} = \frac{X_i - \mathbb{E}(X_i)}{\sqrt{nVar(X_i)}} \sim \mathcal{N}\left(0, \frac{1}{n}\right), i = 1, 2, \ldots, n. \tag{14}$$

3.4 Multivariate Gaussian error linear unit

Trabelsi et al. [37] present a split-type ReLU-based activation function $\mathbb{C}ReLU$, which applies ReLU [1] on both of the real and the imaginary part of a complex neuron:

$$\mathbb{C}ReLU(z) = ReLU(\Re(z)) + iReLU(\Im(z)). \tag{15}$$

At the same time, $\mathbb{C}ReLU$ inherits the shortcoming of ReLU, i.e., nonsmooth and less-probabilistic. Motivated by GELU [16], we present a multivariate Gaussian Error Linear Unit (*MGELU*) and propose a complex variant $\mathbb{C}GELU$ of it. Given an n-dimensional standard normal random vector X, i.e., $X_i \sim \mathcal{N}(0, 1)$ and $X_i \perp\!\!\!\perp X_j$, we define the *MGELU* as:

$$\begin{aligned} MGELU(X) &= X\prod_i^n P(X_i \le x_i) \\ &= X\prod_i^n \Phi(x_i) = \frac{1}{2^n}X\prod_i^n\left(1 + erf\left(\frac{x_i}{\sqrt{2}}\right)\right). \end{aligned} \tag{16}$$

From (11)(12)(13)(16), we can formulate $\mathbb{C}GELU$ as:

$$\mathbb{C}GELU(z) = z * \frac{1}{4}(1 + erf(x))(1 + erf(y)). \tag{17}$$

4 Experiments

In this section, we present empirical results on the oriented object detection in aerial images dataset DOTA-v2.0 [9], which consists of a large number of small objects. We perform a sanity check of our approach to demonstrate its effectiveness and analysis its performance in anti-translation, as shown in Table 6 on page 13.

4.1 Datasets and settings

Dataset. DOTA-v2.0 [9] is a common large-scale dataset of oriented object detection in aerial images, which contains 11, 268 images varied from sensors and platforms, with ground sample distance (GSD) range from $0.1m$ to $4.5m$ per pixel. Moreover, DOTA-v2.0 annotates 1, 793, 658 oriented object instances in orientations, scales, and shapes. The object instances annotated with an oriented bounding box (OBB) comprise 18 common categories. To avoid the problem of overfitting, DOTA-v2.0 provides the training, validation, and test sets. The proportion of the training and validation sets is smaller than that of the test set. The training set consists of 1, 830 images and 268, 627 annotations, while the validation set contains 593 images and 81, 048 annotations. Furthermore, DOTA-v2.0 releases two test subsets, namely test-dev and test-challenge, without annotations. The test-dev set

Table 1 Results of DOTA-v2.0 test-dev set

	Methods	Backbone	AP_{50}	AP_{75}	mAP
Anchor-free	H2RBox-v2	R50	51.3	25.1	27.4
	H2RBox-v2	R101	51.9	25.8	28.2
	Ori-RepPoints	R50	53.9	31.1	31.3
	Ori-RepPoints	R101	54.3	31.7	31.5
Anchor-based	ReDet	Re50	56.7	35.4	33.9
	RoI Trans	R50	55.2	34.4	33.2
	RoI Trans	R101	55.8	35.1	33.9
	RoI Trans	Swin-T	58.8	35.5	34.5
	H2RBox-v2 (Ours)	R50	$52.1^{+0.8}$	$26.5^{+1.4}$	$28.4^{+1.0}$
	Ori-RepPoints (ours)	R50	$54.7^{+0.8}$	$32.0^{+0.9}$	$32.0^{+0.8}$
	ReDet (ours)	Re50	$56.8^{+0.1}$	$36.2^{+0.8}$	$34.4^{+0.5}$
	RoI Trans (ours)	R50	$56.0^{+0.8}$	$35.6^{+1.2}$	$34.2^{+1.0}$
	RoI Trans (ours)	R101	$56.4^{+0.6}$	$36.4^{+1.3}$	$34.6^{+0.7}$
	RoI Trans (ours)	Swin-T	**59.2** $^{+0.4}$	**36.6** $^{+1.1}$	**35.3** $^{+0.8}$

Table 2 Comparison with different normalization and activation functions

$\mathbb{C}BN$	$\mathbb{C}LN$	$\mathbb{C}GN$	$\mathbb{C}ReLU$	$\mathbb{C}GELU$	AP_{50}
✓			✓		61.7
✓				✓	61.9
	✓		✓		62.3
	✓			✓	61.8
		✓	✓		62.7
		✓		✓	63.1

contains 2, 792 images. The test-challenge set contains 6, 053 images. However, the test-challenge set is available only during the challenge. Hence, we conduct two experiment settings. First, we train the model on the training set and evaluate it on the validation set for ablation and case studies. Second, for a fair comparison, we train the model on the union of the training and validation set and evaluate it on the test-dev set as majorities do.

Settings. Following the standard data processing pipeline in MMRotate [52], we crop the high-resolution images into 1024×1024 patches with an overlap of 200 pixels for training and inference. During testing, we merge the detection results of all patches for evaluation. We conduct all experiments on a single NVIDIA RTX3090 GPU. Following H2RBox-v2 [51], all the models are optimized with AdamW [26] with an initial learning rate of 5e-5 and a weight decay of 0.05 unless otherwise stated. We initialize all the models with ImageNet-1K [7] pretrain weight and fine tune 12 epochs totally. The mini-batch size is 2. Besides, a linear learning rate warmup schedule is adopted for 500 initial iterations, and the learning rate is divided by ten after Epoch 8 and Epoch 11. To avoid overfitting, we

Table 3 Effects of real SPE and complex SPE

Method	Backbone	Real SPE	Complex SPE		Channels	Flops (G)	Params (M)	AP_{50}
			Complex Enc.	Real Dec.				
RoI trans	R50					25.3	55.4	60.9
	R101					33.1	74.4	61.4
	R50	✓			256	47.7	65.6	62.1
				✓	256	58.0	70.3	61.7
		✓		✓	128	52.6	67.9	61.0
			✓	✓	128	52.2	65.4	63.1
		✓	✓	✓	128	62.8	70.2	62.7

Table 4 Anti-translation test of RoI trans with ResNet50

AP_{50}		Δx					Avg.
		– 256	– 64	0	64	256	
Δy	– 256	$76.8^{-3.6}$	$77.4^{-3.0}$	$77.1^{-3.3}$	$75.2^{-5.2}$	$77.7^{-2.7}$	$76.8^{-3.6}$
	– 64	$76.5^{-3.9}$	$77.8^{-2.6}$	$78.4^{-2.0}$	$75.5^{-4.9}$	$78.9^{-1.5}$	$77.4^{-3.0}$
	0	$80.2^{-0.2}$	$80.3^{-0.1}$	80.4	$78.3^{-2.1}$	$80.0^{-0.4}$	$79.8^{-0.6}$
	64	$77.4^{-3.0}$	$77.3^{-3.1}$	$77.7^{-2.7}$	$75.0^{-5.4}$	$75.3^{-5.1}$	$76.5^{-3.9}$
	256	$77.3^{-2.6}$	$78.1^{-2.3}$	$80.2^{-0.2}$	$74.9^{-5.5}$	$78.7^{-1.7}$	$77.8^{-2.6}$
Avg.		$77.6^{-2.8}$	$78.2^{-2.2}$	$78.8^{-1.6}$	$75.8^{-4.6}$	$78.1^{-2.3}$	$77.7^{-2.7}$

Table 5 Anti-translation test of RoI trans with ResNet101

AP_{50}		Δx					Avg.
		– 256	– 64	0	64	256	
Δy	– 256	$79.6^{-2.1}$	$79.9^{-1.8}$	$80.7^{-1.0}$	$77.4^{-4.3}$	$79.4^{-2.3}$	$79.4^{-2.3}$
	– 64	$78.2^{-3.5}$	$78.5^{-3.2}$	$80.0^{-1.7}$	$76.3^{-5.4}$	$79.8^{-1.9}$	$78.6^{-3.1}$
	0	$80.5^{-1.2}$	$81.9^{+0.2}$	81.7	$78.4^{-3.3}$	$81.7^{+0.0}$	$80.8^{-0.9}$
	64	$78.6^{-3.1}$	$78.9^{-2.8}$	$79.8^{-1.9}$	$75.4^{-6.3}$	$77.8^{-3.9}$	$78.1^{-3.6}$
	256	$80.2^{-1.5}$	$81.0^{-0.7}$	$82.0^{+0.3}$	$77.2^{-4.5}$	$80.5^{-1.2}$	$80.2^{-1.5}$
Avg.		$79.4^{-2.3}$	$80.0^{-1.7}$	$80.8^{-0.9}$	$76.9^{-4.8}$	$79.8^{-1.9}$	$79.4^{-2.3}$

randomly flip the images through the horizontal, vertical, and diagonal directions with a total probability of 0.75.

4.2 Results

The results in Table 1 on page 11 demonstrate our approach outperforms the traditional oriented object detection frameworks, including anchor-free and anchor-based methods. Taking the state-of-the-art anchor-based detector RoI Trans [8] for example, our approach achieves an accuracy AP_{75} of 1.2% higher with the backbone ResNet50 [15]. When the original RoI Trans takes a stronger feature extractor ResNet101 as the backbone, our approach with ResNet50 also gives a gain of 0.5% in AP_{75}. The same phenomenons are observed in anchor-free detector Ori-RepPoints [23] and weakly supervised detector H2RBox-v2 [51]. Our approach achieves an accuracy of 0.9% and 1.4% higher in the same backbone, and 0.3% and 0.7% higher compared with the stronger backbone.

4.3 Ablation study

Effects of normalization and activation functions. In Table 2 on page 12, we compare the different normalization and activation functions in the complex encoder. For a fair comparison, we use its real variant in the real decoder. Combined with different activation functions, CGN achieves better performance than CBN [37] and CLN, when we train with a small batchsize. On the other side, $\mathbb{C}GELU$ achieves a competitive performance with $\mathbb{C}ReLU$ The results show that the combination of CGN and CGELU achieves the best performance with 63.1% AP_{50} on DOTA-v2.0 val set.

Effects of real SPE and complex SPE. To control the model size, we change the Pie Channels in the middle convolution layer. As shown in Table 3 on page 12, only utilizing the real SPE, it achieves 62.1% AP_{50}, while it gives a 63.1% AP_{50} when adopting complex SPE with the complex encoder and real decoder. However, the combination of real SPE and complex SPE only achieves 62.7% without a performance boost.

Effects of complex encoder. To mitigate the effect of model size, we deepen the real decoder with two convolution layers for testing the effect of the proposed complex encoder. The results in Table 3 on page 12 indicate the complex encoder gives a performance boost with 1.4%, while the deepened real decoder only achieves 61.7% AP_{50}.

Effects of anti-translation. CNNs are regarded as translation equivariant. To test the capability of our approach to encoding the spatial information, we design the anti-

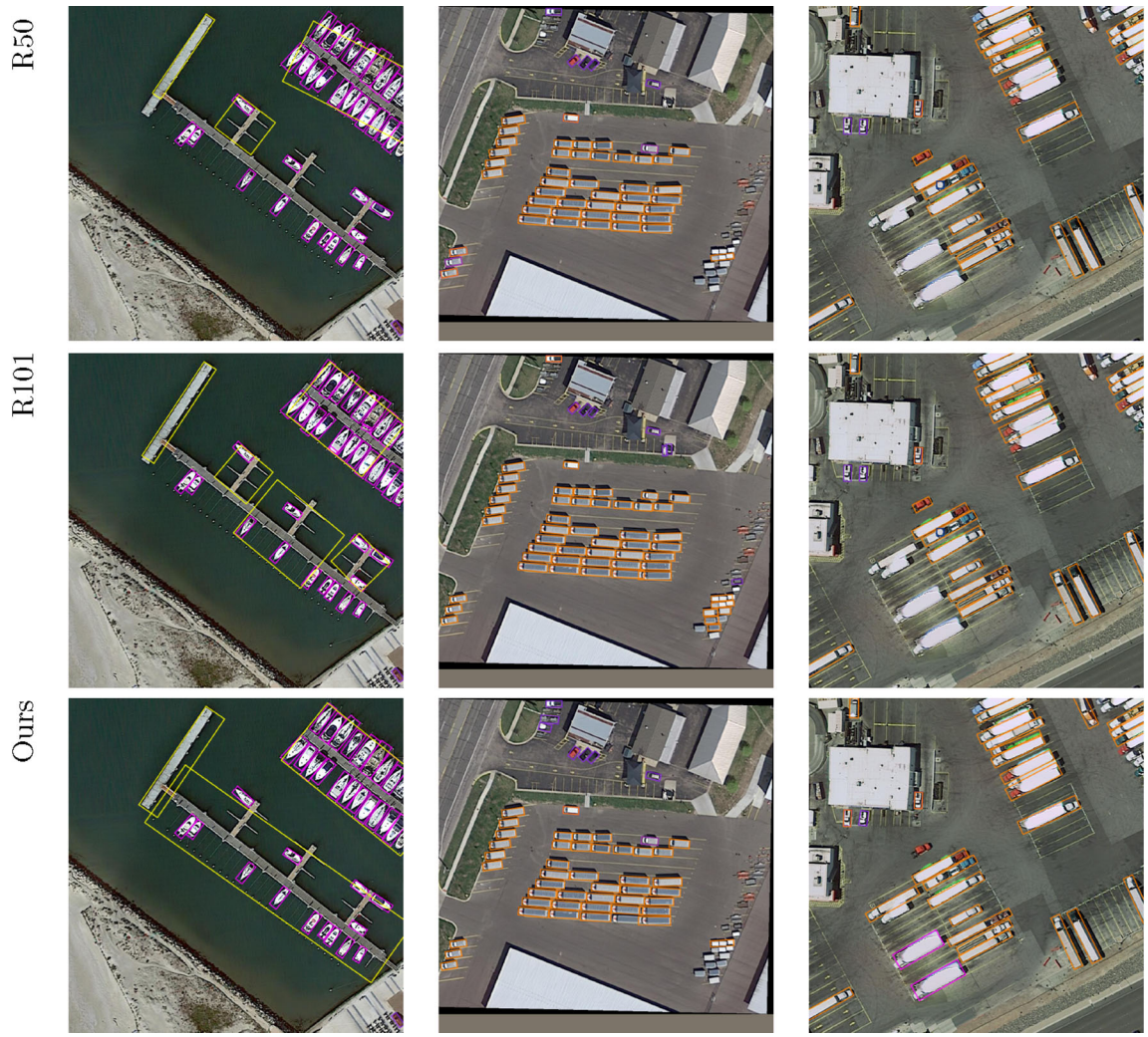


Fig. 3 Qualitative comparison

translation experiments by translating the input images along X and Y directions with ± 64 and ± 256 pixels. We overfit the models on the train val set and test on the validation set. Table 4 on page 13 presents the results of the baseline RoI Trans [8] and Table 6 on page 13 shows the results of our method. The baseline achieves an average accuracy of 77.7%, which is 2.7% lower than the non-translation test. Our method gives an average accuracy of 80.4%, which is 1.7% lower than the nontranslation test. It indicates our PieF can capture more spatial information and achieve better translation equivariance. In Table 5 on page 13, we also present the anti-translation test of RoI Trans [8] with a stronger backbone, e.g., ResNet101 [15]. It achieves an average accuracy of 79.4%, which is 2.3% lower than the non-translation test. Although our approach utilizes the lightweight backbone ResNet50, it achieves better performance than the stronger one.

4.4 Qualitative results

Figure 3 on page 14 presents the qualitative results between our methods and the baseline RoI Trans [8] equipped with different backbones, i.e., R50 and R101 [15]. The first row presents the qualitative results of RoI Trans equipped with R50, and the second row is the results of R101. The results of our proposed methods are drawn in the last row. The visual results demonstrate that our methods can predict more accurate rotations and scales of 2D rotated bounding boxes. With the comparison in the third column, our methods have a higher recall and successfully detect blurred objects. In Fig. 4 on page 15 and Fig. 5 on page 16, we present more visual results of the baseline RoI Trans and our methods, which indicates our methods outperform the baseline with accurate rotations and classifications in small object detection.



Fig. 4 More qualitative results of RoI Trans with R50

5 Discussion

Computation overhead is an essential problem for CVNNs compared with their real counterpart. As shown in Table 3 on page 12, the computational FLOPs of our approach is as twice as the baseline with a similar model size. Moreover, our method can achieve better average performance in anti-rotation test but not have better anti-rotation ability (Table 6).

6 Conclusion

In this paper, we explore the ability of CVNNs to explicitly encode the spatial information and propose the position information encoding fields (PieF), which can be seamlessly integrated into the existing pipelines. To train the PieF, we present the complex variants of multivariate Gaussian normalization and multivariate Gaussian error linear unit. The basic idea is applicable to hypercomplex processing. We demonstrate the effectiveness of our method on DOTA-v2.0, which consists of a large number of small objects. The results indicate that our PieF has a



Fig. 5 More qualitative results of our methods

Table 6 Anti-translation test of RoI trans (ours) with ResNet50

AP_{50}		Δx					Avg.
		– 256	– 64	0	64	256	
Δy	– 256	$78.4^{-3.7}$	$79.4^{-2.7}$	$81.5^{-0.6}$	$80.5^{-1.6}$	$80.2^{-1.9}$	$80.0^{-2.1}$
	– 64	$77.2^{-4.9}$	$78.3^{-3.8}$	$80.6^{-1.5}$	$79.8^{-2.3}$	$81.1^{-1.0}$	$79.4^{-2.7}$
	0	$82.4^{+0.3}$	$82.8^{+0.7}$	82.1	$80.7^{-1.4}$	$83.0^{+0.9}$	$82.2^{+0.1}$
	64	$79.4^{-2.7}$	$80.3^{-1.8}$	$81.2^{-0.9}$	$78.3^{-3.8}$	$79.0^{-3.1}$	$79.6^{-2.5}$
	256	$78.4^{-3.7}$	$81.6^{-0.5}$	$83.1^{+1.0}$	$79.8^{-2.3}$	$80.9^{-1.2}$	$80.8^{-1.3}$
Avg.		$79.2^{-2.9}$	$80.5^{-1.6}$	$81.7^{-0.4}$	$79.8^{-2.3}$	$80.8^{-1.3}$	$80.4^{-1.7}$

stronger ability to capture position information and anti-translation. However, our method has poor performance in computation overhead and anti-rotation tests. In the future, we will go on to explore the novel architecture in CVNNs to reduce the computation overhead and enhance the anti-rotation ability.

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Data availability The data support the findings of this study are "DOTA v2.0" openly available in https://captain-whu.github.io/DOTA/dataset.html.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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